

to receive US FDA approval after being discovered and developed in-house,” Khorakiwala tells *BT*. Wockhardt’s decision to invest in discovering the drug came at a time when antibiotic research had lost momentum and many large drugmakers were pulling back from the field.

Developing a new antibiotic requires years of investment and involves high risk as the outcome is uncertain. Also, hospitals and physicians restrict the use of new drugs to slow antimicrobial resistance. That is why AstraZeneca sold its antibiotics business to Pfizer in 2016, Novartis exited antibacterial research in 2018, and Sanofi transferred its anti-infectives arm to Evotec. Others redirected resources towards oncology, immunology and rare diseases where potential profits are much higher.

Wockhardt saw the same landscape and drew the opposite conclusion. “It became clear that large pharmaceutical companies were stepping away from antibiotic research. But new antibiotics are still needed because bacteria continue to develop resistance,” says Khorakiwala.

The company invested approximately \$800 million, roughly ₹6,800 crore, in antibiotic research and development over two decades. The programme, led by Dr. Mahesh Patel, Chief Scientific Mentor, and Sachin Bhagwat, Chief Scientist for Drug Discovery, who co-invented the zidebactam-cefepime combination, produced six novel drug candidates after testing thousands of molecules on multiple bacterial targets. Three have already been commercialised: Zaynich, Emrok and Miquaf. The remaining three, Odrate, WCK 4282 and Foviscu, are at various stages of clinical development.

## THE TROUBLED YEARS

The commercial significance of Zaynich is inseparable from the difficult path Wockhardt took to bring it to the market. A series of acquisitions between 2003 and 2009 led to a massive debt burden and, finally, corporate debt restructuring. Then, in 2013, US FDA import alerts shut down the company’s ability to export to its most lucrative market. For years, income from domestic formulations, insulin and the UK business serviced debt, while the antibiotic research programme continued in the background.

“Establishing a drug discovery programme takes considerable

they typically address one or two organisms, whereas Zaynich covers the entire Gram-negative space,” says Khorakiwala. “It kills bacteria rapidly and comprehensively.”

The Phase 3 ENHANCE-1 trial, conducted across 530 patients in 64 hospitals and research centres in the US, Europe, Latin America, China and India, tested Zaynich head-to-head against meropenem, one of the most widely used heavy-duty antibiotics in clinical practice. Zaynich achieved clinical recovery and infection elimination in 89% of patients, as against 68.4% for meropenem.

Drug-resistant infections were

## THE DEBT REDUCTION

• Wockhardt has reduced debt despite the investments in antibiotic research

₹3,213  
CR (₹DEBT  
IN FY20)

₹1,859  
CR (₹DEBT  
IN FY25)

time,” says Khorakiwala. “It takes about ten years just to build such a programme across approximately 15 different scientific disciplines. A further seven to ten years is a normal timeline for any pharmaceutical company to bring a drug to approval.”

Zaynich belongs to a new class of antibiotics; new antibiotic classes emerge once in around 20 years. It is designed to treat serious multidrug-resistant Gram-negative bacterial infections, which are becoming increasingly untreatable in hospital settings as existing drugs have lost their potency. The drug works through a dual-action mechanism.

“Competing products exist, but

associated with nearly 4.95 million deaths globally in 2019, according to the landmark GRAM study published in *The Lancet*, a figure researchers say has worsened since. The US alone reports more than 2.8 million antimicrobial-resistant infections annually, resulting in over 35,000 deaths, according to the CDC. The FDA had signalled its view of Zaynich’s importance well before approval, granting it Priority Review, Fast Track designation and Qualified Infectious Disease Product status, the last of which also confers ten years of additional market exclusivity after approval, which delays generic competition and extends the revenue runway.

According to the 2026 Antimi-